

Write a Java program to demonstrate the relationship I1 extends I2, I3. Use constructors, method overriding and method overloading to calculate the total and average marks of a student.

Steps

1. Create interface `Total`.
2. Declare a method to calculate total marks.
3. Create interface `Average`.
4. Declare a method to calculate average marks.
5. Create interface `Result`.
6. Make `Result` extend both `Total` and `Average`.
7. Create class `Student`.
8. Make `Student` implement `Result`.
9. Declare marks of three subjects.
10. Create a constructor to initialize the marks.
11. Implement the methods inherited from both interfaces.
12. Calculate total marks.
13. Calculate average marks.
14. Overload the calculation method.
15. Create a `Student` object in `main()`.
16. Call the methods.
17. Display total and average.

Algorithm

START

1. Create interface `Total`.
2. Declare total calculation method.
3. Create interface `Average`.

4. Declare average calculation method.
5. Create interface Result extending Total and Average.
6. Create class Student implementing Result.
7. Declare marks M1, M2 and M3.
8. Initialize the marks using constructor.
9. Implement the Total method.
10. Calculate Total = M1 + M2 + M3.
11. Implement the Average method.
12. Calculate Average = Total / 3.
13. Overload the calculation method.
14. Create Student object in main().
15. Call the methods.
16. Display Total and Average.

STOP

```
PS C:\Users\venik\OneDrive\Assignmets\Assigment2\I1extendsI2,I3> & 'C:\Pro  
ges' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\f941a  
1ExtendsI2I3'  
Total: 245  
Average: 81.66666666666667  
Sum of two marks: 155  
PS C:\Users\venik\OneDrive\Assignmets\Assigment2\I1extendsI2,I3>
```